Muhammad Abdullah Haroon

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Lahore, Pakistan

Education

National University of Computer and Emerging Sciences (FAST-NUCES) — **Lahore, Pakistan**
Bachelor of Science in Computer Science (BSCS) — *Aug 2023 – Expected 2027*
FYP: Multi-Domain, Bi-Script Fake News Detection & Deepfake Audio Detection System for Urdu Language

Work Experience

FAST School of Computing — **Jul 2026 – Sep 2026**
Research Intern — *Lahore, Pakistan*

- Extended an independently-established Urdu FND finding (a length-confound shortcut driving **macro-F1 collapse from 0.771 to 0.005**) by adding English FND and sarcasm-detection domains under a unified zero-shot transfer protocol (**XLM-RoBERTa** vs. TF-IDF + LR/SVM), under the supervision of Dr. Maryam Bashir.
- Built and curated a **33-dataset, 288,899-example** Urdu benchmark across sentiment, abusive language, fake news, and QA-validation tasks; ran a deduplication pass removing **4,196** train-test leaks.
- Benchmarked **5 models** (XLM-RoBERTa, mBERT, Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B) under a baseline-guarded relative-robustness metric, surfacing a 3-tier task hierarchy from **84.5–97.3%** (sentiment) down to **48.1–64.4%** (fake news / hate speech).

QuickFuel — **Jun 2026 – Jul 2026**
IT Intern — *Lahore, Pakistan*

- First person in company history – not just among interns – to introduce **AI/ML** work; spoke up and stood out in a traditionally .NET/WPF-focused engineering team, auditing the Management Portal and identifying **9 AI integration opportunities**.
- Built a sentiment analysis prototype (**Python, LLMs, FastAPI, PostgreSQL**) and a semantic similarity search system on **ChromaDB** for context-aware product retrieval in place of keyword matching.

Optimus Automate — **Jun 2026 – Jul 2026**
AI Automation Intern — *Remote*

- Built an **LLM-powered** email triage and response agent (**Gmail API** integration, intent classification, auto-drafted replies) and a **RAG-based** document Q&A/summarization tool for PDFs, grounding every answer in retrieved source content to avoid hallucinated responses.

Research

Domain Robustness of Multilingual NLP Models Across Urdu and Roman Urdu Scripts
M. A. Haroon, M. Bashir | *In preparation – final stages; targeting Springer Language Resources and Evaluation (Q1)*

- Released **33 Urdu/Roman Urdu** text-classification datasets (**288,899** labeled examples) across sentiment, abusive language, fake news, and QA-validation tasks, publicly released on Hugging Face, Zenodo (DOI: 10.5281/zenodo.22195610), and GitHub.
- Evaluated **5 models** (XLM-RoBERTa, mBERT, Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B) under a baseline-guarded robustness protocol, establishing a 3-tier task hierarchy: **84.5–97.3%** relative robustness on sentiment vs. **48.1–64.4%** on fake news/hate speech.

When Benchmarks Mislead: Shortcut Learning, Length Confounds, and the Limits of Cross-Dataset Generalization in Multilingual Fake News and Sarcasm De-

tection
M. A. Haroon | arXiv:2607.14131

- Identified a length confound in the Ax-to-Grind Urdu dataset (fake articles average **3.4**× longer than real), driving a transfer collapse to **macro-F1 = 0.005** in one direction against **0.771** in reverse; reproduced on English FND (WELFake vs. ISOT) and sarcasm detection (TweetEval Irony vs. Sarcasm Corpus V2).

Bitcoin Price Direction Prediction via Regime-Aware Multi-Modal Fusion of Social Sentiment and Technical Features
M. A. Haroon | arXiv:2607.23370

- Proposed **Regime-Aware Multi-Modal Learning (RAML)**, conditioning the fusion of FinBERT-scored Reddit sentiment and BiLSTM-encoded OHLCV features on a rolling 24-hour volatility regime; evaluated on **3,491** hourly observations, reaching **macro-F1 of 0.5474 (3h) / 0.5513 (6h)**.

Projects

TaxChronos | Python, LangChain, Qdrant, Gemini API, RAGAS GitHub
Multilingual RAG system for Pakistani tax queries with English–Urdu code-switching

- Built a dense-retrieval RAG pipeline (**Qdrant** + **Gemini**) that answers tax queries in **under 20 seconds** versus 2–3 minutes of manual document search; evaluated on a code-switched English–Urdu test set against baseline **RAGAS** scores (Faithfulness, Context Precision, Context Recall).

Daily Digest | FastAPI, PostgreSQL (Neon), Gemini, JWT, Docker, Resend GitHub
AI news aggregator delivering personalized, scheduled email digests – live in production

- Deployed a Generative AI product live in production with **15+ active users**, running an unattended fetch → rank → summarize → email pipeline on free-tier infrastructure, with a custom Pakistan-relevance ranking function and **Gemini**-generated summaries.

Urdu Domain-Robustness Benchmark | XLM-RoBERTa, BERT, Llama, Qwen, Mistral GitHub
Open-source benchmark repository and evaluation tooling backing the Domain Robustness research above

- Engineered the full data pipeline behind the **33-dataset, 8-task**, dual-script benchmark, and open-sourced **33 datasets** plus **5** fine-tuned/evaluated NLP models across Hugging Face, Zenodo, and GitHub.

Bilingual WhisperBench | Python, OpenAI Whisper, FastAPI, SQLAlchemy GitHub
Zero-shot benchmarking dashboard for compact Whisper models on code-mixed speech

- Ran a zero-shot evaluation of **Whisper Tiny/Base/Small** (39M–244M params) on **327** curated Urdu-English code-mixed recordings from **11** bilingual speakers; introduced the **Switch Point Error Rate (SPER)** metric and identified Whisper Small + English-forced inference as the best configuration (**WER 0.899**).

Technical Skills

Languages: Python, C, C++, C#, SQL, JavaScript, TypeScript, HTML, CSS
Machine Learning & Generative AI: PyTorch, TensorFlow, Keras, Scikit-learn, NLP, LLMs, Transformers, LangChain, RAG, Ollama, Prompt Engineering, OpenAI Whisper, LSTM, n8n
Databases: PostgreSQL, MS SQL Server, SQLite, MongoDB, Qdrant, ChromaDB, FAISS
Backend, Frontend & Tooling: MERN(Node.js, Express.js, MongoDB, React.js), FastAPI, SQLAlchemy, REST APIs, .NET, Tailwind CSS, Git/GitHub, Docker, Render, Jupyter/Colab, Claude Code, Codex, Antigravity, Cursor

Certifications & Leadership

AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents – (Udemy) | **Machine Learning with Python** – (IBM)
Data Analysis and Visualization with Python – (Microsoft) | **Introduction to Generative AI** – (Google)

Team Leadership & Scrum Master – **FYP**/semester teams: sprint planning, presentations, task ownership. **Competitions & Campus Involvement** – **SQL Query Master Competition** SOFTEC'25 & **Media Officer**; **ACM Speed Programming**; **MS Office instructor & delivered AI Workshops**, Pact College; **Deputy Head of PR**, FAST Business Club.